

Change the slope

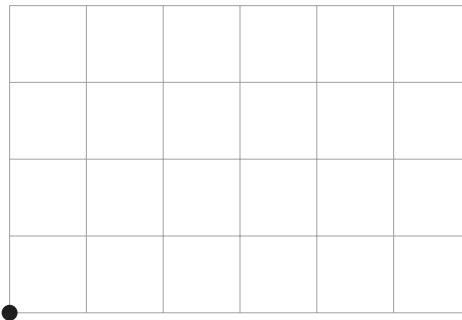
Name: _____

Date: _____

Use a UNIT SQUARE. Start at $(0, 0)$ heading up-right with slope $s = \text{vertical distance} / \text{horizontal distance}$. Unfold reflections into the straight line $y = sx$. Stop at a corner.

1. Rational directions.

Trace unfolded slopes $2/3$ and $3/2$. If $s = p/q$ with positive coprime integers p, q , why is the first lattice point (q, p) ? Predict the ending corner and count bounces.



each little square is a whole reflected table

2. A path that never reaches a corner.

Take $s = \sqrt{2}$. If the unfolded path reached an integer point (u, v) with $u > 0$, what would v/u equal? Why is that impossible?

A proof, not just a long drawing:

3. Never hitting is not the same as going everywhere.

Start instead at $(0, 1/2)$ with slope 1. Trace four bounces. Does this path hit a corner? Does it explore the whole square?

A theorem beyond this page: The $\sqrt{2}$ -slope path launched from $(0, 0)$ above is dense: it comes arbitrarily close to every point. Your no-corner argument alone does *not* prove this. Unfolding leads to straight-line motion on a torus, a subject in dynamical systems.

Try it with objects. Explain by pointing, talking, or drawing.